

Small Virtual Office Network Design and Simulation

Student Name: Abraham Cerritos

Course: CSCE 3530.401

Date: July 11, 2026

Project Overview

This project involved designing and testing a small virtual office network using Cisco Packet Tracer. The network included one router, one Ethernet switch, and three client computers named PC0, PC1, and PC2. The devices were connected so that the computers could communicate through the switch and router. Both IPv4 and IPv6 addressing were configured and tested.

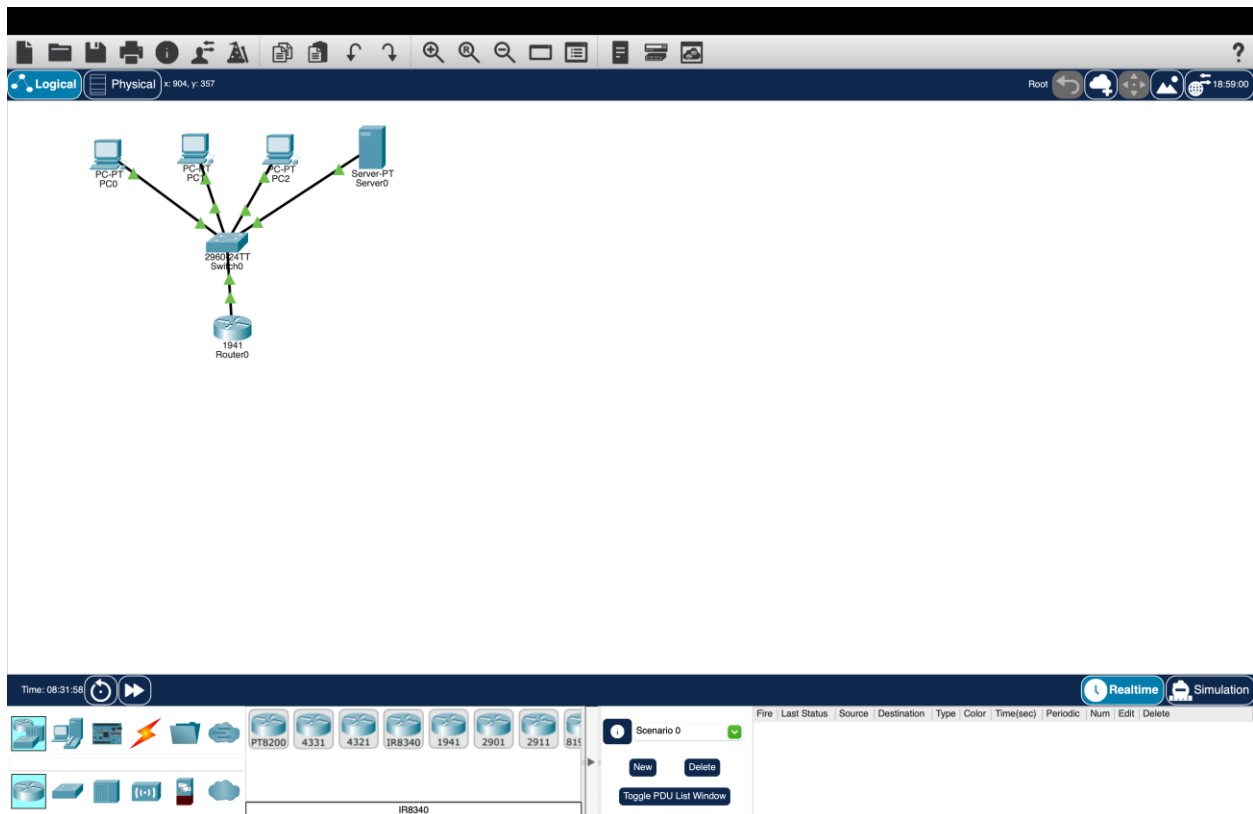
Objectives of the Project

The main objectives of this project were to create a small office network in Cisco Packet Tracer, connect a router, switch, and three computers, configure IPv4 and IPv6 addresses, configure the correct default gateways, test communication between the computers, capture network traffic, and demonstrate that the network was working correctly.

Network Design Explanation

The network was designed using a simple star topology. PC0, PC1, and PC2 were connected to the Ethernet switch. The switch was then connected to the router. The switch allowed the computers to communicate on the local network, while the router acted as the default gateway.

Each computer was assigned an IPv4 address and an IPv6 address. The addresses were placed in the same network so the devices could communicate with each other. The router interface was also assigned to IPv4 and IPv6 addresses.



Networking Protocols Used

The Ethernet was used to connect the computers, switch, and router. It allowed frames to move between devices on the local network.

IPv4 was used to assign addresses to the router and computers. It allowed the devices to identify and communicate with each other.

IPv6 addresses were also configured to demonstrate communication using the newer Internet Protocol addressing system.

ARP was used to match IPv4 addresses with MAC addresses. This allowed devices on the local network to find the correct destination hardware address.

ICMP was used during the ping tests. It sent echo requests and echo replies to confirm that the computers could communicate.

ICMPv6 was used when testing the IPv6 connections. It performed a similar function to ICMP but was used with IPv6 addresses.

Device Configurations

The router was connected to the Ethernet switch and configured with IPv4 and IPv6 addresses. The router address was used as the default gateway for the client computers.

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Global Settings

Display Name Router0

Hostname Router

NVRAM

Erase

Save

Startup Config

Load...

Export...

Running Config

Export...

Merge...

Device Clock: 05:25:23 Sun Mar 7 1993 UTC

Equivalent IOS Commands

A summary of U.S. laws governing Cisco cryptographic products may be found at:
<http://www.cisco.com/wai/aeaport/crypto/tool/stgty.html>
If you require further assistance please contact us by sending email to report@cscom.com.

Cisco C1801S41/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID F815240206
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Top

The Ethernet switch is connected to PC0, PC1, PC2, and the router. It forwarded Ethernet frames between the connected devices.

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

Global Settings

Display Name

Switch0

Hostname

Switch

Serial Number

Serial Number

NVRAM

Erase

Save

Startup Config

Load...

Export...

Running Config

Export...

Merge...

Device Clock

05:26:10 Sun Mar 7 1993 UTC

Equivalent IOS Commands

SLINEPROTO-5-OPDMN: Line protocol on Interface FastEthernet0/1, changed state to up

SLINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

SLINEPROTO-5-OPDMN: Line protocol on Interface FastEthernet0/2, changed state to up

SLINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

SLINEPROTO-5-OPDMN: Line protocol on Interface FastEthernet0/3, changed state to up

SLINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

SLINEPROTO-5-OPDMN: Line protocol on Interface FastEthernet0/4, changed state to up

SLINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

SLINEPROTO-5-OPDMN: Line protocol on Interface FastEthernet0/5, changed state to up

Top

PC0 was configured with an IPv4 address, subnet mask, default gateway, and IPv6 address.

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC0

Interfaces FastEthernet0

Gateway/DNS IPv4

DHCP

Static

Default Gateway 0.0.0.0

DNS Server 192.168.1.100

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

Device Clock: 05:26:37 Sun Mar 7 1993 UTC

NTP

Server Ip:

Authentication

Enable

Disable

Password:

PTP

Profile

Power

Enable

Disable

Top

PC1 was configured with an IPv4 address, subnet mask, default gateway, and IPv6 address.

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC1

Interfaces FastEthernet0

Gateway/DNS IPv4

DHCP

Static

Default Gateway 0.0.0.0

DNS Server 192.168.1.100

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

Device Clock: 05:26:59 Sun Mar 7 1993 UTC

NTP

Server Ip:

Authentication

Enable

Disable

Password:

PTP

Profile

Enable

Disable

Power

Top

PC2 was configured with an IPv4 address, subnet mask, default gateway, and IPv6 address.

Physical
Config
Desktop
Programming
Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC2

Interfaces FastEthernet0

Gateway/DNS IPv4

☒ DHCP
☐ Static

Default Gateway 0.0.0.0

DNS Server 192.168.1.100

Gateway/DNS IPv6

☐ Automatic
☒ Static

Default Gateway

DNS Server

Device Clock: 05:27:49 Sun Mar 7 1993 UTC

NTP

Server Ip:

Authentication

☐ Enable
☒ Disable

Password:

PTP

Profile

Power

☐ Enable
☒ Disable

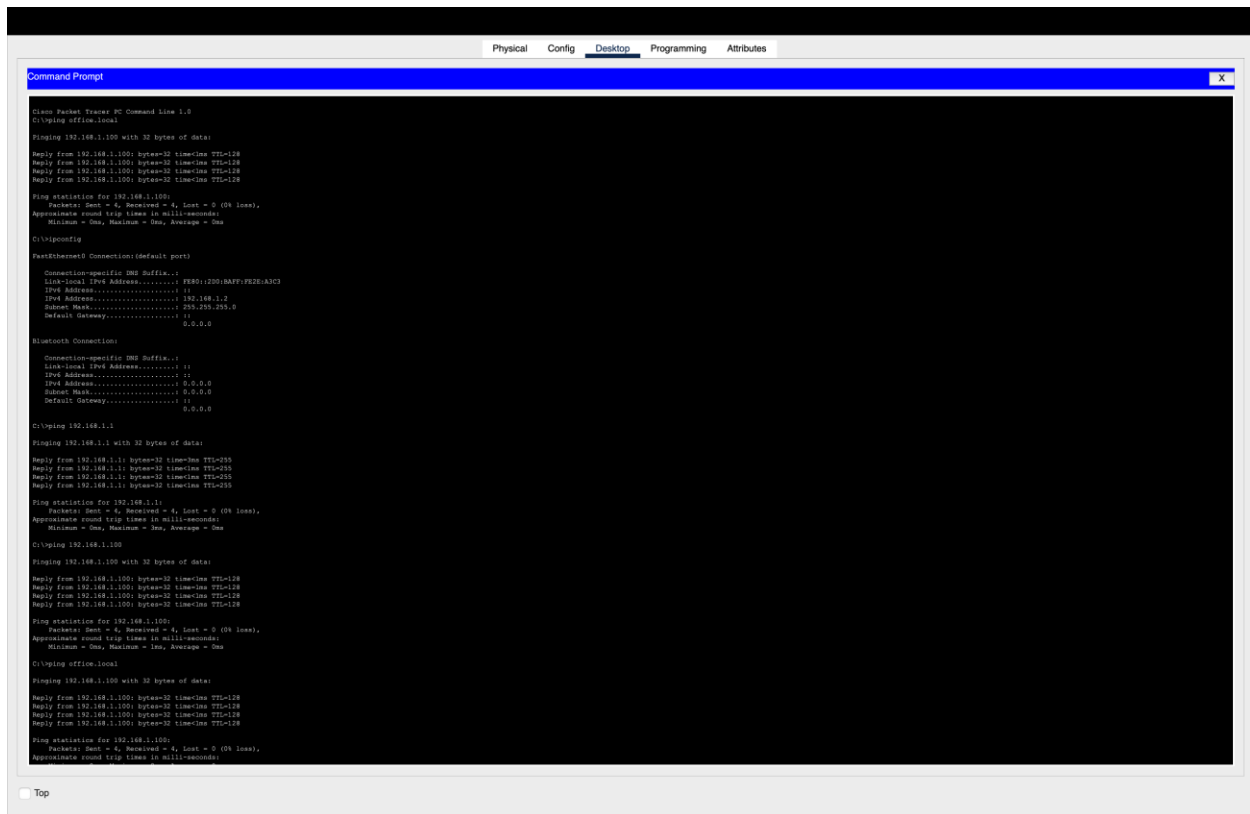
Top

Network Testing and Results

The network was tested by sending ping messages between PC0, PC1, and PC2. IPv4 ping tests were completed first. The successful replies showed that the IPv4 addresses, subnet masks, connections, and gateways were configured correctly.

IPv6 ping tests were also completed. The successful replies showed that the devices could communicate using IPv6.

Packet captures were used to observe the traffic created during tests. The captures showed ICMP echo requests and echo replies for IPv4. They also showed ICMPv6 traffic during the IPv6 tests.



```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping office.local

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv4 Address...: FE80::2D0:BAFF:FE2E:A3C7
    IPv4 Address...: 192.168.1.2
    Subnet Mask...: 255.255.255.0
    Default Gateway...: 192.168.1.1

Loopback0 Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv4 Address...: 
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: 0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping office.local

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

The testing confirmed that all three computers could communicate successfully across the network.

Challenges Faced and Solutions

One challenge was making sure that each device had the correct IP address and default gateway. An incorrect address or gateway could prevent communication. I solved this by checking the configuration of each computer and making sure all devices were using the correct network information.

Another challenge was testing both IPv4 and IPv6. I had to make sure that both address types were configured separately. After reviewing the settings and repeating the ping tests, the devices communicated successfully.

I also had to make sure that every virtual cable was connected to the correct device and interface. I checked the topology and restarted the tests after confirming the connections.

Conclusion

This project helped me understand how a small office network is designed and tested. I learned how routers, switches, and computers work together in a network. I also practiced configuring IPv4 and IPv6 addresses, setting default gateways, and using ping commands to test connectivity. The packet captures helped me see how ICMP and ICMPv6 messages travel between devices. Overall, the network worked correctly, and all three computers were able to communicate.